Data	Suggestions
Cora	GNN: Two-layer stakcing-based GCN. Hyper-parameters: { hidden size:16, dropout ratio:0, learning rate:0.01, weight decay:5e-4 }
Computer	GNN: Two-layer stakcing-based GCN. Hyper-parameters: { 'hidden':256, 'dropout':0.1, 'lr':0.01, 'l2':5e-4,' }
Photo	GNN: Two-layer GCN in which the aggregation results are concatenated to predict the node labels. Hyper-parameters: { 'hidden':128, 'dropout':0.1, 'lr':0.005, 'l2':5e-4, }
Physics	GNN: Two-layer GCN in which the aggregation results are concatenated to predict the node labels. Hyper-parameters: { 'hidden':128, 'dropout':0.1, 'lr':0.005, 'l2':5e-4, }
genius	GNN: Two-layer GraphSAGE with mean aggregator. Hyper-parameters: {'activation': ReLU, hidden size: 128, lr: 0.01, 'l2':5e-4}.
DD	GNN: Three-layer GIN in which the aggregation results are added, and graph representation vector is obtained with Global sum readout operation. Hyper-parameters: { 'hidden':128, 'dropout':0.1, 'lr':0.01, 'l2':5e-4 }.
PROTEINS	GNN: Two-layer stakcing-based GIN, and graph representation vector is obtained with Global sum readout operation. Hyper-parameters: { 'hidden':32, 'dropout':0, 'lr':0.01, 'l2':5e-4 }.
NCI1	GNN: Three-layer GIN in which the aggregation results are added, and graph representation vector is obtained with Global sum readout operation. Hyper-parameters: { 'hidden':64, 'dropout':0.0, 'lr':0.01, 'l2':5e-4 }.
NCI109	GNN: Three-layer GIN in which the aggregation results are added, and graph representation vector is obtained with Global sum readout operation. Hyper-parameters: { 'hidden':128, 'dropout':0.1, 'lr':0.01, 'l2':5e-4 }
Epsonion	Three components are used in the GNN. In each component, two-layer stakcing-based GCN are employed, and these components are concatenated to obtain the node representations. The user and item rep- resentations are first concatenated or summed up, and then fed into an MLP for prediction
Amazon-Sports	GNN: Two-layer GAT in which the aggregation results are concatenated to formulate the final node representations. The Hadamard operation is adopted when calculating the messages. The final prediction is obtained based on the dot product of the user and item representations.